Team KnowlAgents

CosmicCompass

Subject Matter AI Expert in K12 Astronomy - Solar System and Universe

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Overview

Domain Definition

Domain Choice: K12 Astronomy - Solar System and Universe

This is a complex, fact-heavy scientific domain that requires precise data and benefits significantly from multi-modal (text + image) explanations.

Core Capabilities of SME:

- Multi-Modal Q&A: Retrieval of text and diagrams from technical corpora.
- Content Generation: Creation of Quizzes, Reports (PDF/DOCX), and Presentations (Beamer).
- Task Automation: Direct delivery of materials via Email.

Data Collection and Organisation

Corpus: Curated collection of authoritative Astronomy textbooks and academic materials.

Supported Formats:

- PDF & PPTX: For rich layout and image extraction.
- DOCX & Markdown: For structured text content.

Organization:

- Metadata-Driven: JSON manifests link source documents to their constituent chunks and extracted images.
- Centralized Assets: All processed assets stored in a structured outputs/ directory.

Preprocessing and Chunking

Cleaning, deduplication, chunking strategy, and reasoning

01

Pipeline: process.py handles ingestion, cleaning (preprocess_text), and segmentation.

02

Strategy: Hierarchical Chunking.

- Child Chunks (256 tokens): Small, focused segments for precise retrieval.
- Parent Chunks (1024 tokens): Larger context windows provided to the LLM to ensure semantic continuity.

03

Reasoning: Prevents loss of context typical in small chunks while maintaining the search precision of granular segments.

Embedding and Indexing

- **Embedding Model:** clip-ViT-B-32
  - **Choice:** A Multi-Modal model capable of embedding both Text and Images into a shared vector space, enabling seamless image retrieval.

- **Vector Database**: Elasticsearch
  - **Schema**: Hybrid index mapping storing dense vectors (dense_vector) alongside full text fields.

- **Reranking (Bonus)**: Implemented Cross-Encoder (ms-marco-MiniLM-L-6-v2) to rescore top-k results for higher accuracy.

RAG Retrieval Workflow

- **Hybrid Search Approach:**
  - Keyword Search (BM25): Captures exact term matches (e.g., specific star names).
  - Semantic Search (k-NN): Captures conceptual matches via embeddings.

- **Retrieval Logic:**
  - Retrieve Top-20 from both methods → Deduplicate → Rerank → Return Top-5 to Agent.

- **Multi-Modal Capability:**
  - Explicit logic to retrieve a minimum number of images (min_images) when visual aids are requested.

System Architecture

Modular Design:

- core/: Agent logic, prompts, and middleware.
- rag/: Indexing and retrieval logic.
- api/: FastAPI backend and HTML frontend.
- tools/: Tools required by the agents.

API Design:

- Decoupled endpoints for Chat (start task) and Resume (handle HITL decision), enabling a stateless yet interruptible workflow.

Agent Workflow and Tools

ARCHITECTURE

Supervisor-Worker Pattern (LangGraph)

- Supervisor: Plans workflow and routes tasks (e.g., "Generate PDF" → "Email User").
- Workers: Specialized agents for Content Generation and Emailing
- We use deepseek-v3 model for running the agents.

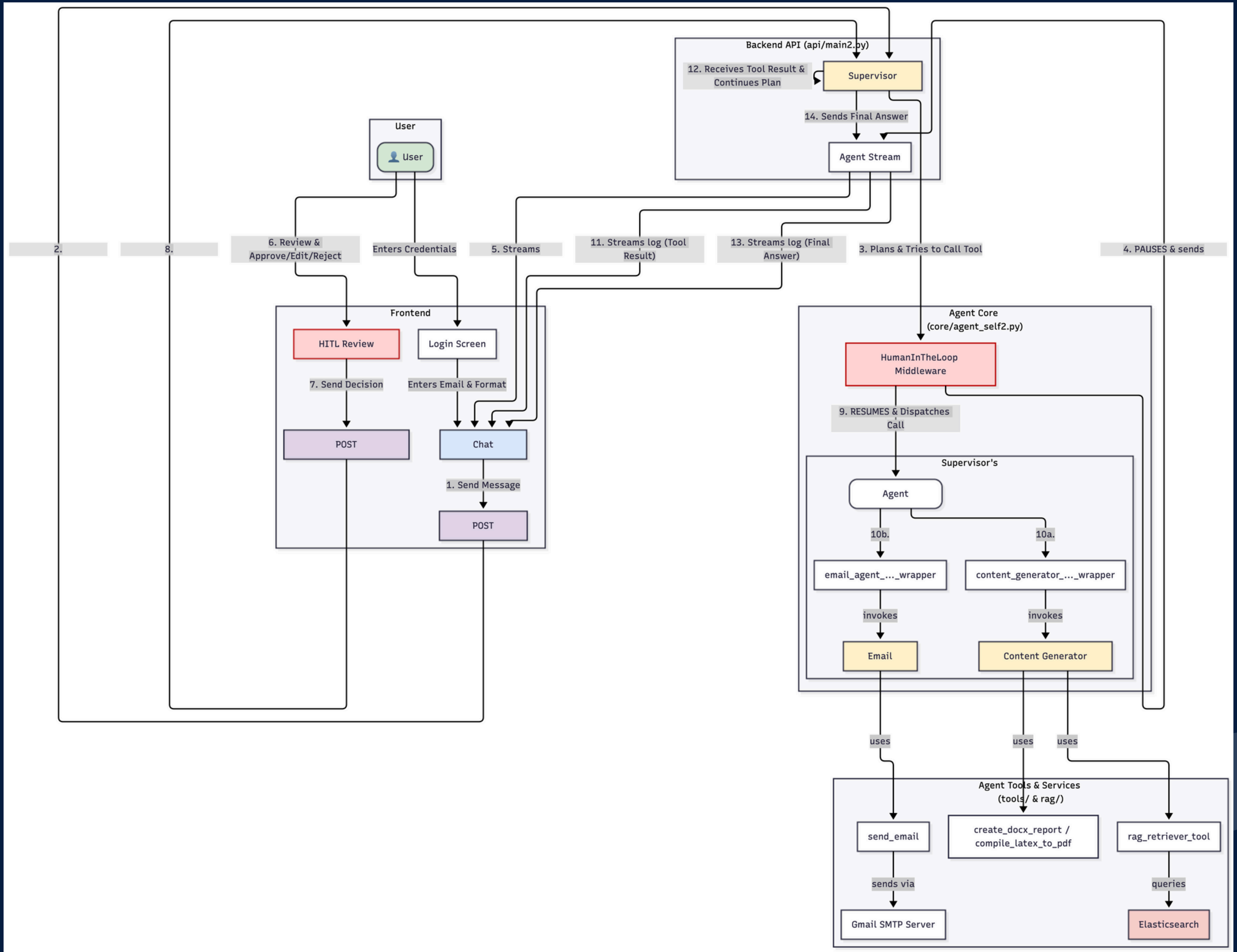
ROBUSTNESS

- **Memory:**
  - InMemorySaver maintains state across interrupts.
- **Retries:**
  - ToolRetryMiddleware handles transient failures.

ROUTING

Dynamic tool selection based on conversation history and user intent.

Agent Workflow Diagram



Prompting Strategy and Iteration

Evolution:

- Moved from naive zero-shot prompts to structured "Chain of Thought" instructions.

Successful Instructions:

- One-Shot Prompting:
  - Providing a full LaTeX/Beamer template in the prompt was critical for successful presentation generation.
- Zero-Shot Prompting:
  - Sufficient for simpler DOCX generation.
- Context Injection:
  - Injecting User Email and Format Preference at the session start prevented logic errors.

Document Generation and Email

- **PDF/Slides:** Automated compilation using pdflatex. Supports Beamer class for creating slide decks from text descriptions.
- **DOCX:** Automated report generation with embedded images using python-docx.
- **Email:** Integrated SMTP client (smtplib) for sending generated files as attachments directly to the user.

Human-in-the-Loop (HITL) & Guardrails

We implemented Human-in-the-Loop verification. Since autonomous agents can hallucinate or send emails to the wrong people, our system pauses execution before any sensitive action. A modal pops up, allowing the user to edit the email draft or the file topic. The agent only resumes once the human approves.

Human in the Loop

- **Mechanism**: Graph Interrupts via LangGraph.
- **Workflow**: System pauses before sensitive actions (Email/File Gen) → User reviews/edits in UI Modal → System resumes.

Guardrails

- **PII Middleware**: Masks sensitive data (credit cards, keys).
- **Content Filter**: Deterministic blocking of harmful queries.

Conclusion

Summary

Successfully delivered a robust, agentic SME capable of complex reasoning and autonomous content creation.

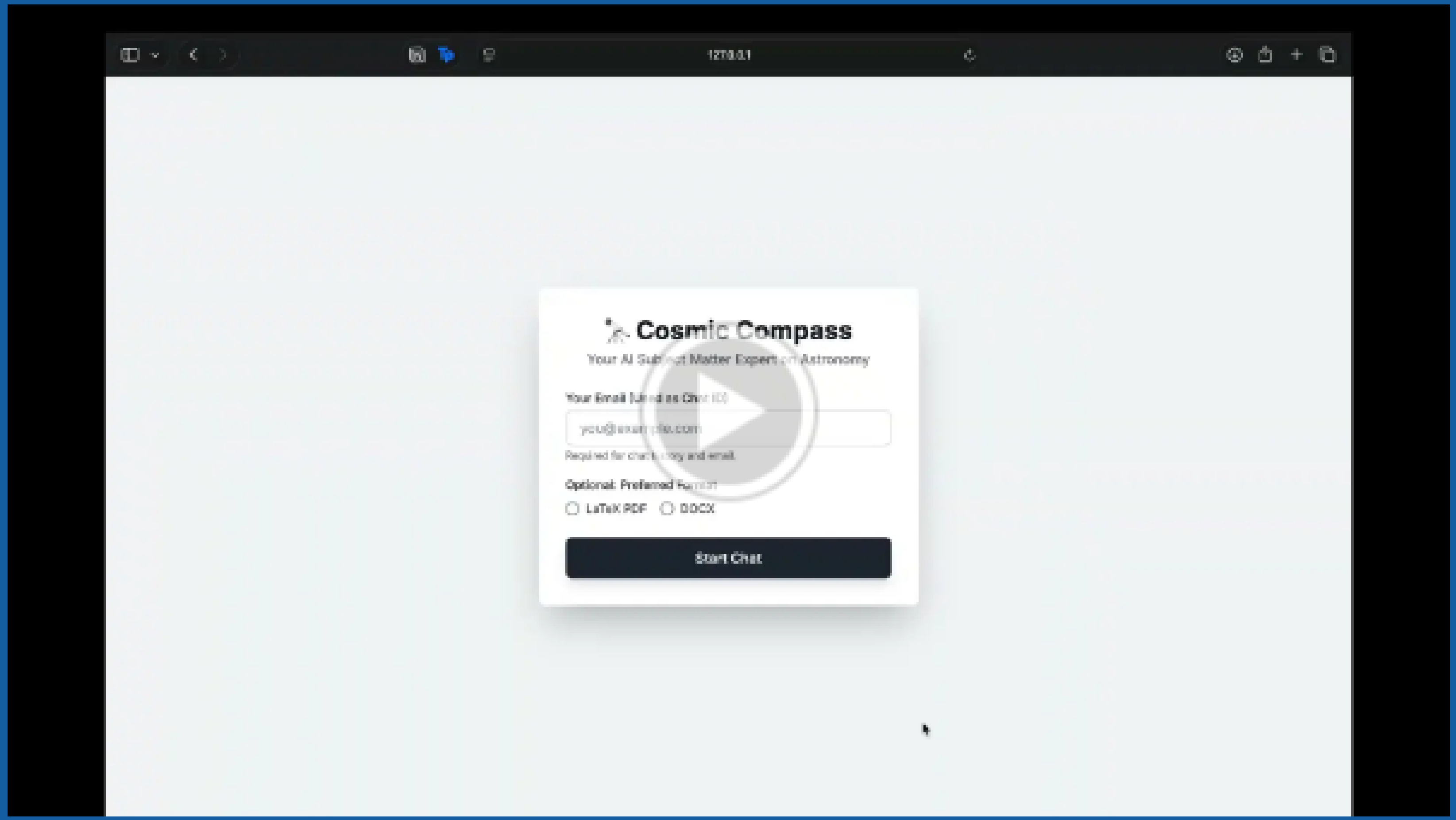
Key Achievements

Seamless Hybrid RAG, Multi-Modal Handling, and Robust Human-in-the-Loop verification.

Future Work

Persistent Database Storage

DEMO



THANK YOU!